

INTRODUCTION

Cavity-nesting bird species such as **Eastern bluebirds**, wood ducks, chickadees, purple martins, American kestrels, and barn owls play an important role in maintaining ecological balance across the Eastern Coast and the DMV region. These birds help **control insects**, **regulate rodent populations**, and **support biodiversity**.

Urbanization, deforestation, agricultural expansion, and wetland drainage have reduced the number of natural nesting cavities, including hollow trees and standing dead trees (snags). Many cavity-nesting species struggle with **lower breeding success**, **greater competition from invasive species**, and **increased exposure to predators**.

Artificial nest boxes are commonly used in conservation, but many are **poorly designed**. Some **lack proper ventilation, drainage, or predator protection, and overheating**. Most conservation efforts still rely on **one-size-fits-all** designs that do not account for species-specific needs or environmental monitoring.

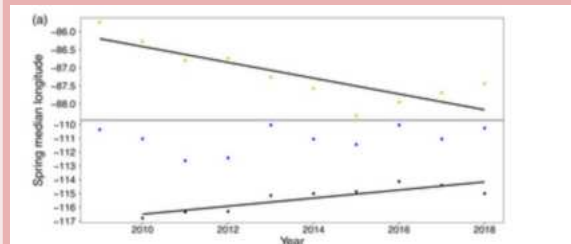


Table 1. Sonnleitner, Jared, et al. Changes in the spring (a) and fall (b) median longitude during migration of eastern (black points), western (yellow points), and mountain bluebirds (blue points) from daily population centroid longitudes and latitudes predicted using generalized additive models. ESA Journals, <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs2.4316>.

OBJECTIVE

This project addresses these limitations by developing **smart, modular nesting pods** for cavity-nesting birds in Frying Pan Farm Park. By integrating **species-specific dimensions, sustainable materials, and environmental sensors**, this project aims to **improve occupancy and breeding rate** while collecting environmental data to **support conservation efforts**.

SOURCES

Scott-Baumann, J. F., Morgan, E. R., & Cogan, T. A. (2022). Effects of nest-box environment on fledgling success rate and pathogen load. *Parasitology*, 149(9), 1186–1192. <https://doi.org/10.1017/S0033182022000695>
May, H. L. (2001, January). Artificial nesting structures - NRCS field office technical guide. https://efotg.scegov.usda.gov/references/public/SC/Nesting_Structures.pdf
Stumpf, Katie (2024) "TEMPERATURE AFFECTS NEST BOX SELECTION, NEST SUCCESS, AND NESTLING SIZE IN A SOUTHEASTERN POPULATION OF EASTERN BLUEBIRDS," Georgia Journal of Science, Vol. 82, No. 2, Article 7.

COLLECTING NEST DATA WITH SENSOR-EQUIPPED PODS TO SUPPORT EAST COAST BIRD CONSERVATION

METHODS

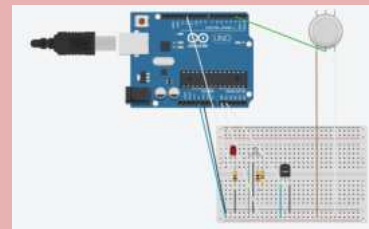
Physical Design

- Nesting pods were constructed using **HDPE** to compare material performance.
- Species-specific dimensions for **Eastern bluebirds**.
- Each pod included:
 - Ventilation slits beneath the roof
 - Drainage holes
 - Predator pole baffles
 - Natural internal nesting materials (hemp fiber)



Programming & Testing

- Sensors were programmed using **Arduino IDE** to record environmental data.
- Data were exported into **Google Sheets** for analysis

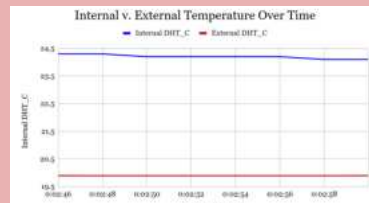


Sensor Integration

- Arduino Uno microcontroller
- A solar-powered battery
- Temperature sensors (**TMP36, DHT22**)
- Humidity sensors (**DHT22**)
- Air quality sensors (**MQ135**)

Field Deployment

- Nest boxes were installed 4–5 ft high
- UV-resistant coatings** reduced overheating
- Baseline data collected before bird occupancy



0:02:40	1.25	74.51	166.12	19.9	32.5	14
0:02:42	1.24	74.02	165.24	19.9	32.7	14
0:02:44	1.23	73.05	163.48	19.9	32.6	14
0:02:46	1.24	74.02	165.24	19.9	32.6	14
0:02:48	1.24	74.02	165.24	19.9	32.6	14
0:02:50	1.23	73.05	163.48	19.9	32.6	14
0:02:52	1.24	74.02	165.24	19.9	32.6	14
0:02:54	1.24	74.02	165.24	19.9	32.7	14
0:02:56	1.23	73.05	163.48	19.9	32.7	14
0:02:58	1.25	74.51	166.12	19.9	32.6	14

Time (HH:MM:SS)	TMP36_Voltage	TMP36_C	TMP36_F	DHT_C	DHT_Humidity	MQ135_PPM
3:06:02	1.49	98.93	201.07	nan	nan	14
4:06:02	1.46	96	204.79	nan	nan	14
5:06:02	1.47	96.97	198.64	nan	nan	14
6:06:02	1.46	96	203.04	nan	nan	14
7:06:02	1.43	92.58	207.43	nan	nan	14
8:06:02	1.45	95.02	203.04	nan	nan	14

DATA

Observations

- TMP36: Fairly steady readings across testing
- DHT: Temperature and humidity columns are all nan (sensor did not return valid data)
- MQ135: Stayed at 14 for every reading (unusual)

Analysis

- Stable temperature means design helped regulate heat
- DHT sensor was likely not connected correctly
- MQ135 did not reflect real air-quality

CONCLUSIONS



Success

- Eastern bluebirds** inhabited and nested in pods
- Nesting box maintained **stable internal temperature**
- Internal flooding** was prevented
- HDPE material showed **high durability** and **weather resistance**

- UV coating **minimized overheating** (no major temperature spikes)
- Nesting activity indicates potential for **improved breeding success**

Shortcomings

- Drainage holes collected moisture
- Solar panel could not support the internal camera and Arduino
- Predator baffles efficiency unknown

EXTENSIONS

- Use **more environmentally friendly construction methods**
- Improve drainage system and finer fibers to **prevent moisture buildup**
- Upgrade solar panel capacity** to reliably power sensors and camera
- Test and refine predator baffles** to ensure effectiveness
- Increase number of deployed pods** to compare occupancy rates across locations
- Integrate **more advanced sensors** for long-term climate tracking